

Model Building

Gen AI rules

- ▶ You are allowed to use Gen AI to improve your text but
- ▶ You must submit both the original and the improved text
- ▶ You cannot use Gen AI to generate your code or interpret your results.
- ▶ I run the code for the group projects and if there are any packages used that I don't use then this will generate errors and you will drop marks and I will know you used Gen AI
- ▶ Last year someone dropped 30% of their marks for using it and not declaring it
- ▶ I use e.g. GPTzero to see if Gen AI has been used and if I suspect it has then I will ask you to come in and have an oral exam about aspects of your report.

Questions part 1

1. When should we use R squared? When should we use adjusted R squared?
2. When should we use residual standard error?
3. When should we use VIF? When should we use a correlation matrix?
4. What's the difference between dependence and a confounding variable? Is there a dependence relationship between a confounder and the predictor it influences. What should we do with it?

Questions part 2

5. When should we merge two levels of a categorical variable?
Just by looking at the box plot? How do you use a box plot?
6. When should we use ANOVA?
7. When should we use subsets regression? When should we use interactions?
8. Why should we use F test? Why can't we simply use `display(lm)` and assess each predictor individually?
9. If the ANOVA test for a categorical variable is not significant, can we just drop out the predictor?
10. Which part should we include in exploratory data analysis?
Which parts are in analysis?

Guide to model building

- ▶ First off: statistical modelling is something of an art.
- ▶ If you give the same (complex) data set to two statisticians you may find that they come up with two different models.
- ▶ If they know what they are doing they won't be very different but they will be differences – so there is not necessarily one right answer.
- ▶ You must learn to use your judgement (and interactions with team members when working together) to make decisions about what steps to take in an analysis.
- ▶ So you must be able to defend any steps you take.
- ▶ This means you cannot analyse the data without thinking carefully about which steps to take.
- ▶ If you are not thinking about what the results mean at every step and are focussing only on diagnostics or plots then you are not doing it correctly!

First steps

- ▶ When faced with a new data set and a specific research question a statistician will do the following before starting to run regressions:
 - ▶ Think about what relationships you expect there to be between the variables (i.e. you expect age to be positively correlated with salary)
 - ▶ but be prepared to reconsider your expectations when more variables come into play i.e. there may be confounding/non-linearity

First thoughts

- ▶ Think carefully about whether there may be omitted predictors i.e. predictors you would like to have observed but haven't.
- ▶ Is it possible to obtain appropriate data on these from another source, do the experts have these data?
- ▶ If it is not possible to obtain data on these confounders then how may omitting them from the analysis bias the results?
- ▶ If you are working with an expert then make sure you know what they think should be going on
 - ▶ beware though of assuming that the expert will be completely correct as they may have pre-conceptions

First actions

- ▶ Look at the summary of the data
- ▶ Check that you know which variables are what (type, role)
- ▶ Plot those that are most relevant (typically the outcome vs the predictors)
- ▶ Check these plots make sense (are they non-linear, do the signs make sense?)
- ▶ Are there potential outliers? No need to do anything yet but bear them in mind.
- ▶ I cannot emphasise how important it is to have thought hard about the problem and the available data BEFORE you start running regressions.

Analysis plan part 1:

- ▶ Start the analyses and do approximately this (not necessarily but often in this order) :
 1. Start off with a regression with all the variables and remove those that are non-significant, starting with the one that is least significant (backward elimination) OR start off with the predictors that “make sense” and add all the others, keeping those that are significant (forward selection).
 2. It might be necessary to deal with multicollinearity at this early stage and decisions have to be made about missingness at this point as well.
 3. Sometimes it makes sense to combine predictors in a score (as was done with the Study.Skills)

Analysis plan part 2

4. Transform variables where this makes sense (e.g. if you want to be able to interpret the intercept or if there is clear evidence of non-linearity, heteroscedasticity)
 5. If after an initial regression some predictors have large effects (check the standardised coefficients) check if interactions are also important
 6. Check overall model diagnostics and run the residual plots
 7. Identify outliers (as detailed in WorkShop 5)
 8. Run some cross-validation
- Note: There is typically no need to perform steps 7 and 8 for every single model.

Variable selection

- ▶ At various points you will have to decide whether to include a predictor or not.
- ▶ Do the following:
- ▶ If a predictor is borderline non-significant at the 5% level but the sign is correct leave it initially (see below for when to remove it)
- ▶ If a predictor is non-significant at the 5% level and the sign is not correct remove it
- ▶ If a predictor is significant at the 5% level and does not have the expected sign then think hard! Why is this happening? Are there omitted/confounding variables? Have you made a mistake? If you think there are consider that your expectations may have been wrong and that you need to re-think.
- ▶ If a predictor is significant at the 5% level and has the expected sign then keep them
- ▶ If you have a lot of borderline non-significant predictors (excluding levels of a categorical predictor) remove them.
- ▶ I will never ask you to justify why you removed a non-significant predictor but I will expect justification for non-significant ones

Things that will happen

- ▶ During the course of an analysis you might try different things.

E.g

- ▶ Add and remove predictors
- ▶ Check for multicollinearity
- ▶ Check for significant interactions (MLR only)
- ▶ Combine many-levelled categorical variables
- ▶ Try different transforms (MLR only)
- ▶ Turn continuous variables into categorical variables
- ▶ Remove outliers (MLR only)
- ▶ Run cross-validation (MLR only)

Aim

- ▶ The aim is to get a model or models that are best at predicting and/or best at explaining the relationships between the variables given the data you have and the research question.
- ▶ Demonstrate that you are an honest, capable statistician that can apply modelling techniques well and report back both on the results and limitations of the models.
- ▶ Remember that the best models you can obtain using regression methods may not be very good.
- ▶ I will be judging your analysis not your model

Some rules of thumb

1. A model with parameters that are easier to interpret is often preferable to one that has hard to interpret parameters
2. A model which leads to better predictions is often preferable to one with worse predictions
3. A simple model is often preferable to a more complex model
4. A model with better diagnostics is often preferable to one with worse ones

Some rules of thumb

- ▶ The 4 criteria above are often at odds with one another especially when the improvement is not large (e.g. cross-validation predictions aren't that much better).
- ▶ For the course-work I am happy for you to present 2 models with a preference for one of them.
- ▶ This is not always possible in a working context so decisions need to be made and justified – however you should say that the model is not very good if that is the case.

Missing values

- ▶ Check the EOT instructions for details
- ▶ For categorical predictors with over 10% missing remove missing (with caution)
- ▶ For categorical with over 10% replace missing with level missingness
- ▶ For continuous predictor with over 30% missing remove predictor after looking at MLR or SLR with that predictor to get an idea of impact